

3-Level Geometric Modeler: CSG, Subdivision Trees and Boundary Representation

KAI CAO

New York University
kcao@cs.nyu.edu

Abstract

The problem of computing geometry modeling in 3D is fundamental to computer graphics, path finding, motion planning and numerous branches of computer science. Given a polyhedron P defined by CSG (Constructive Solid Geometry) model, we raise a new idea to present the boundary features (vertices, edges and faces) of P using subdivision method; it has the following steps: 1) define a polyhedron P in CSG model, 2) predicate and subdivide a given box B_0 recursively until all sub-boxes are elementary 3) generate the adequate Subdivision Tree, 4) represent the boundary features based on the Subdivision Tree. Since the great generality and scalability of subdivision method, this idea can also be expanded to support more complex shape with higher order surfaces in 3D.

I. INTRODUCTION

Solid modeling is distinguished from related areas of geometric modeling and computer graphics by its emphasis on physical fidelity. Together, the principles of geometric and solid modeling form the foundation of computer-aided design and in general support the creation, exchange, visualization, animation, interrogation, and annotation of digital models of physical objects.

We describe a polyhedron as three sets : (V, E, F) . V represents the set of vertices of the polyhedron. E represents the set of edges of the polyhedron. F represents the set of faces of the polyhedron.

II. DEFINITIONS

2.1 Regularized set

A set $S \subset \mathbb{R}^3$ is a **regularized set** iff $S = \overline{\text{int}(S)}$.

2.2 Regularized set operation on regularized sets A and B

- $A \cup^* B = \overline{\text{int}(A \cup B)}$
- $A \cap^* B = \overline{\text{int}(A \cap B)}$
- $A -^* B = \overline{\text{int}(A - B)}$
- $A^{c*} B = \overline{\text{int}(A^c)}$

2.3 Solid

A **solid** is defined as a regularized subset of $\mathbb{R}^3$.

III. CSG OBJECTS

A CSG (Constructive Solid Geometry) model describes a polyhedron as regularized set operation on primitives which are the simplest solid objects used for the representation.

3.1 Primitives

Primitives in CSG models are all semi-algebraic sets as follows:

1. **Half Space**: $H = \{(x, y, z) | a * x + b * y + c * z + d \geq 0\}$.
2. **Sphere**: $S = \{(x, y, z) | (x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2 - r^2 \leq 0\}$.
3. **Cylinder**: $C = \{(x, y, z) | A = -x \sin(\theta) + y \cos(\theta) \cos(\phi) + z \cos(\theta) \sin(\phi), B = -y \sin(\phi) + z \cos(\phi), A^2 + B^2 \leq R^2\}$.
4. **Torus**: $T = \{(x, y, z) | (R - \sqrt{x^2 + y^2})^2 + z^2 \leq r^2\}$.

- $B_4 = ([\bar{x}, x_2], [y_1, \bar{y}], [z_1, \bar{z}])$
- $B_5 = ([\bar{x}, x_2], [y_1, \bar{y}], [\bar{z}, z_2])$
- $B_6 = ([\bar{x}, x_2], [\bar{y}, y_2], [z_1, \bar{z}])$
- $B_7 = ([\bar{x}, x_2], [\bar{y}, y_2], [\bar{z}, z_2])$

We name B_i ($i = 0, 1, 2, \dots, 7$) a subbox of B .

3.2 Polyhedron

A **polyhedron** P is a finite solid that is the intersection of half spaces. We define the set containing all polyhedra as $\mathbb{P}$.

IV. SUBDIVISION TREES

4.1 Box

B is a **box** iff $B \subset \mathbb{R}^3$, $B = \{(x, y, z) | x_1 \leq x \leq x_2, y_1 \leq y \leq y_2, z_1 \leq z \leq z_2\}$, which is a regularized subset of $\mathbb{R}^3$.

Another representation of B is $B = ([x_1, x_2], [y_1, y_2], [z_1, z_2])$.

We name the set containing all boxes as $\mathbb{B}$.

4.2 Subdivision on box B

Suppose $B = ([x_1, x_2], [y_1, y_2], [z_1, z_2])$, let $\bar{x} = \frac{x_1 + x_2}{2}$, $\bar{y} = \frac{y_1 + y_2}{2}$, $\bar{z} = \frac{z_1 + z_2}{2}$. The subdivision on B is to subdivide to box B into eight smaller boxes:

- $B_0 = ([x_1, \bar{x}], [y_1, \bar{y}], [z_1, \bar{z}])$
- $B_1 = ([x_1, \bar{x}], [y_1, \bar{y}], [\bar{z}, z_2])$
- $B_2 = ([x_1, \bar{x}], [\bar{y}, y_2], [z_1, \bar{z}])$
- $B_3 = ([x_1, \bar{x}], [\bar{y}, y_2], [\bar{z}, z_2])$

4.3 Features

For a polyhedron $P \subset \mathbb{R}^3$, we use set V , E and F to represent its vertices, edges and faces respectively. The **features** $\Phi(P) = (V, E, F)$.

Then we define the features of a box B corresponding to polyhedron P as follows:

- $\Phi_0^*(B, P) = B \cap V$
- $\Phi_1^*(B, P) = \{(e, e \cap B) | e \in E\}$
- $\Phi_2^*(B, P) = \{(f, f \cap B) | f \in F\}$
- $\Phi^*(B, P) = (\Phi_0^*(B, P), \Phi_1^*(B, P), \Phi_2^*(B, P))$

4.4 Elementary Box

Then we define the **elementary box** B_e for a polyhedron P :

1. **Empty Box**: a box B_e is an empty box iff $B_e \cap P = \emptyset$.
2. **Full Box**: a box B_e is a full box of Polyhedron P iff $B_e \cap P = B_e$.
3. **Vertex Box**: a box B_e is a vertex box iff there exists only one vertex $v \in V \cap B_e$, s.t. $\forall e \in E$, if $e \cap B_e \neq \emptyset$, v bounds e , and $\forall f \in F$, if $f \cap B_e \neq \emptyset$, v bounds F .
4. **Edge Box**: a box B_e is an edge box iff $V \cap B_e = \emptyset$ and there exists only one edge $e \in E$, $e \cap B_e \neq \emptyset$, $\forall f \in F$, if $f \cap B_e \neq \emptyset$, e bounds f .
5. **Face Box**: a box B_e is a face box iff $V \cap B_e = \emptyset$, $E \cap B_e = \emptyset$ and there exists only one face $f \in F$, and $f \cap B_e \neq \emptyset$.

4.5 Definition of S-Tree and AS-Tree

1. Subdivision Tree Node (ST-Node) for box B and polyhedron P , represented as $n(B, P)$: a Subdivision Tree Node is a node contains the following information:

- pointer to the parent of n : pp
- a list of pointers to the eight children of node n : $p_i (i = 0, 1, 2, \dots, 7)$
- a box B
- $\Phi^* (B, P)$

If B is an Elementary Box for P in node $n(B, P)$, then $n(B, P)$ is called an Elementary Subdivision Tree Node (EST-Node). Subdivision on a Subdivision Tree Node $n(B, P)$ is defined as :

- (a) Subdivide box B into 8 subboxes $(B_0, B_1, \dots, B_7)$
- (b) Compute $\Phi^*(B_i, P)(i = 0, 1, 2, \dots, 7)$
- (c) Construct eight ST-Nodes $n(B_i, P)$ and set each pp of $n(B_i, P)$ point to B
- (d) Set p_i of $n(B, P)$ point to $n(B_i, P)$ ($i = 0, 1, 2, \dots, 7$)

2. Subdivision Tree (S-Tree) for box B_s and polyhedron P : A Subdivision Tree (S-Tree) is an octree rooted at $n(B_s, P)$ and in which each node is a Subdivision Tree Node for a box B and polyhedron P . For each inner node $n(B, P)$, the eight children of $n(B, P)$ are $n(B_i, P)$ ($i = 0, 1, 2, \dots, 7$).

3. Adequate Subdivision Tree (AS-Tree) for box B_s and polyhedron P : An Adequate Subdivision Tree(AS-Tree) for box B_s and polyhedron P is a S-Tree whose leaves are EST- Nodes.
We represent an AS-Tree for box B_s and polyhedron P as $AST(B_s, P)$

4. Simplest Adequate Subdivision Tree (SAS-Tree) for a box B_s and polyhedron P :

- (a) Construct a ST-node $n(B_s, P)$ and let it be the root of S-Tree.
- (b) Subdivide $n(B_s, P)$ if it is not an EST-node.
- (c) Keep subdividing every leaf of the S-Tree until it is an AS-Tree, which means that every leaf of the S-Tree is an EST-Node.
- (d) Now the AS-Tree we have is an SAS-Tree. Finally we get an AS-Tree. This tree is an SAS-Tree.
We represent an SAS-Tree constructed by box B_s and polyhedron P as $SAST(B_s, P)$.

4.6 Boolean Operations on AS-Trees

Boolean operations on two adequate subdivision trees $T_1 = AST(B_s, P_1)$ and $T_2 = AST(B_s, P_2)$ rooted at the same box B_s : $T = M(BO, T_1, T_2)$ in which BO is a Boolean operation and T_1 and T_2 are adequate subdivision trees. The result of the function M is an AS-Tree T .

BO are Boolean operations as union ($\cup$), intersection ($\cap$) and difference ($-$).

We also define:

- $T_1 \cup T_2 = M(\cup, T_1, T_2)$
- $T_1 \cap T_2 = M(\cap, T_1, T_2)$
- $T_1 - T_2 = M(-, T_1, T_2)$

The result $T = AST(B_s, P_1 \text{ } BO \text{ } P_2)$.

1. Boolean operation on an elementary box for two polyhedra P_1 and P_2 Suppose box B_e is an elementary box for both P_1 and P_2 . The result of the Boolean operation on an elementary box for two polyhedra P_1 and P_2 is an adequate subdivision tree. We define a function m as: $T = m(BO, B_e, P_1, P_2)$ in which BO is a

Boolean operation as $\cup$, $\cap$ and $-$, P_1 and P_2 are two polyhedra. The result of the function is an adequate subdivision tree. What is more, we define a generalized function m , when BO is a complement, which is also a boolean operation, one of the two polyhedron should be null (because the complement c is a unary operator). Arbitrarily we define $P_2 = Null$, then $T = m(^c, B_s, P_1)$.

P_1 and P_2 are commutative in this function.

(a)

V. BOOLEAN OPERATION ON AN ELEMENTARY BOX B_e FOR TWO POLYHEDRA P_1 AND P_2

5.1 Union

1. If B_e is an empty box of either P_1 or P_2 (let us say P_1), the result $T = m(\cup, B_e, P_1, P_2)$ is a subdivision tree whose root points to B_e and has no children. B_s is an elementary box for P_2 .
2. if B_e is a full box of either P_1 or P_2 (let us say P_1), $T = m(\cup, B_e, P_1, P_2)$ is a subdivision tree whose root points to B_e and has no children. B_s is an elementary box for P_1 .

3. A polygon Box & A polygon Box:

- (a) A polygon Box: if one half plane contains another (a belongs to b)
- (b) A corner Box: if two segments crosses and the cross point lies within the Box (the intersection of a and b is not empty)
- (c) Not a simple Box: if there is no intersection between the two half planes, then the Box will be divided into 3 parts and finally we have to subdivide the Box until we get simple Boxes. (a and b are disjoint)

4. A polygon Box & A corner Box: This is the union of several half planes. Let us say that the half space containing this polygon is p. And the corner c is the intersection of half spaces $s_1, s_2, \dots, s_n$

- (a) if p passes through the end point of corner c: a corner box or a half plane

5. A corner Box & corner Box: Let us suppose that we have corner c_1 and c_2 . c_1 is the intersection of half space $s_1, s_2, \dots, s_n$, while c_2 is the intersection of half space $t_1, t_2, \dots, s_n$.